

Supplementary Material

An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity

Target-Wise Interpretability Atlases

This supplementary file collects the complete target-wise ICSOR raw-head interpretability atlases that are not retained in the main manuscript. Figure 4 of the manuscript is reserved for the COD raw-head Θ_{cc} block only, so the paper can discuss one representative interaction surface without repeating four dense multi-block panels in the main text. Figures S1–S4 provide the full COD, TN, TP, and TSS raw-head atlases.

Each atlas presents the reported-target coefficient blocks obtained from the pre-correction raw-head map $I_{comp}(I_F - \Gamma)^{-1}B$: the scalar bias term b , the first-order operational block W_u , the first-order influent block W_{in} , the operational-curvature block Θ_{uu} , and the operation–influent and influent-curvature blocks Θ_{uc} and Θ_{cc} . The component-space coupling matrix Γ is also shown as a shared structural panel. The ASM component basis is kept complete in every applicable panel, and the heatmaps are drawn from the lower origin so the operational axes begin at HRT and then Aeration. These atlases describe the globally defined raw regression head; the final affine/LP deployment map is piecewise and therefore does not have one global coefficient atlas.

COD raw-head-specific ICSOR coefficient atlas

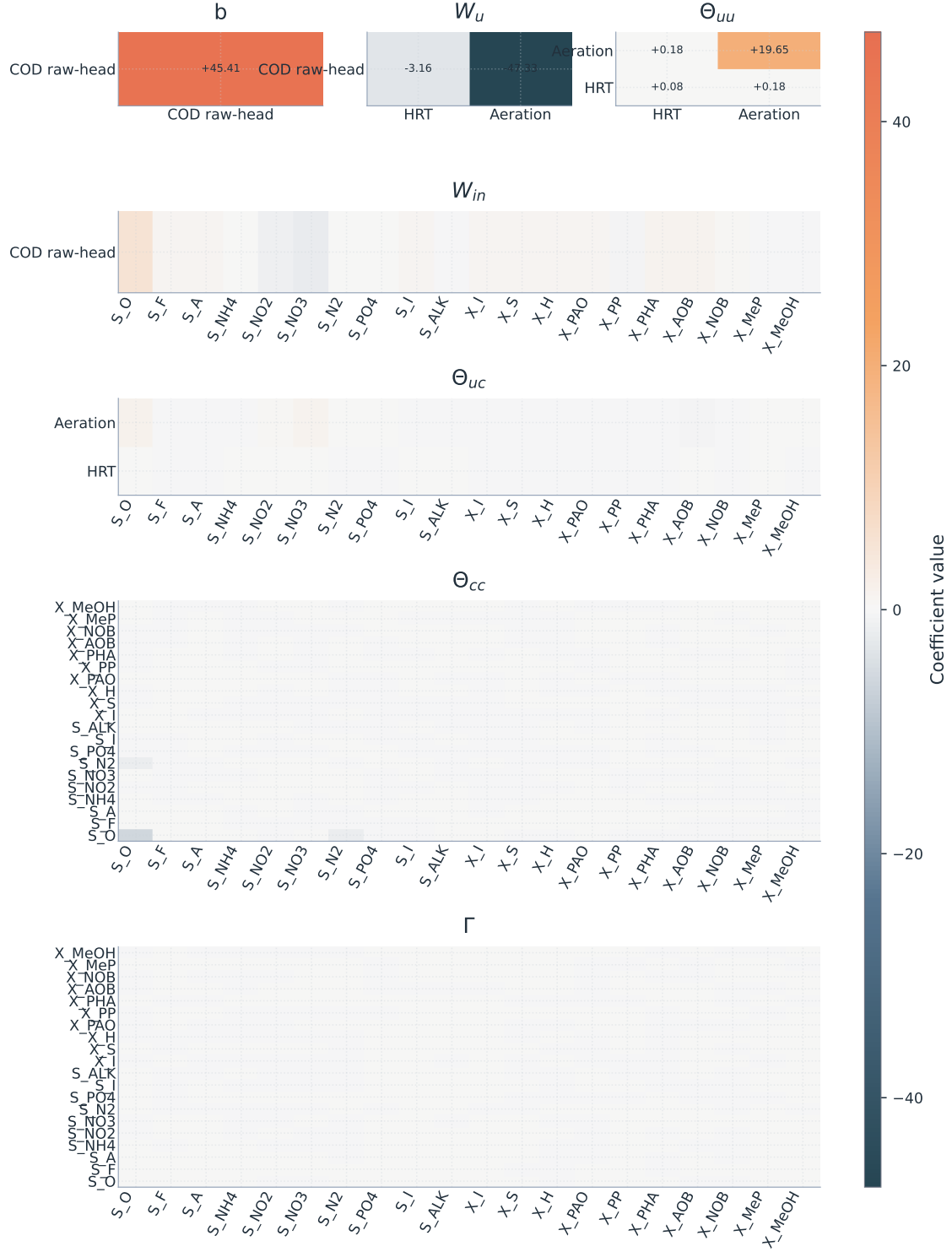


Figure S1: COD raw-head ICSOR coefficient atlas showing the reported-target blocks of $I_{comp}(I_F - \Gamma)^{-1}B$ together with the shared component-space coupling matrix Γ . The symmetric Θ_{uu} and Θ_{cc} panels display mirrored off-diagonal cells.

TN raw-head-specific ICSOR coefficient atlas

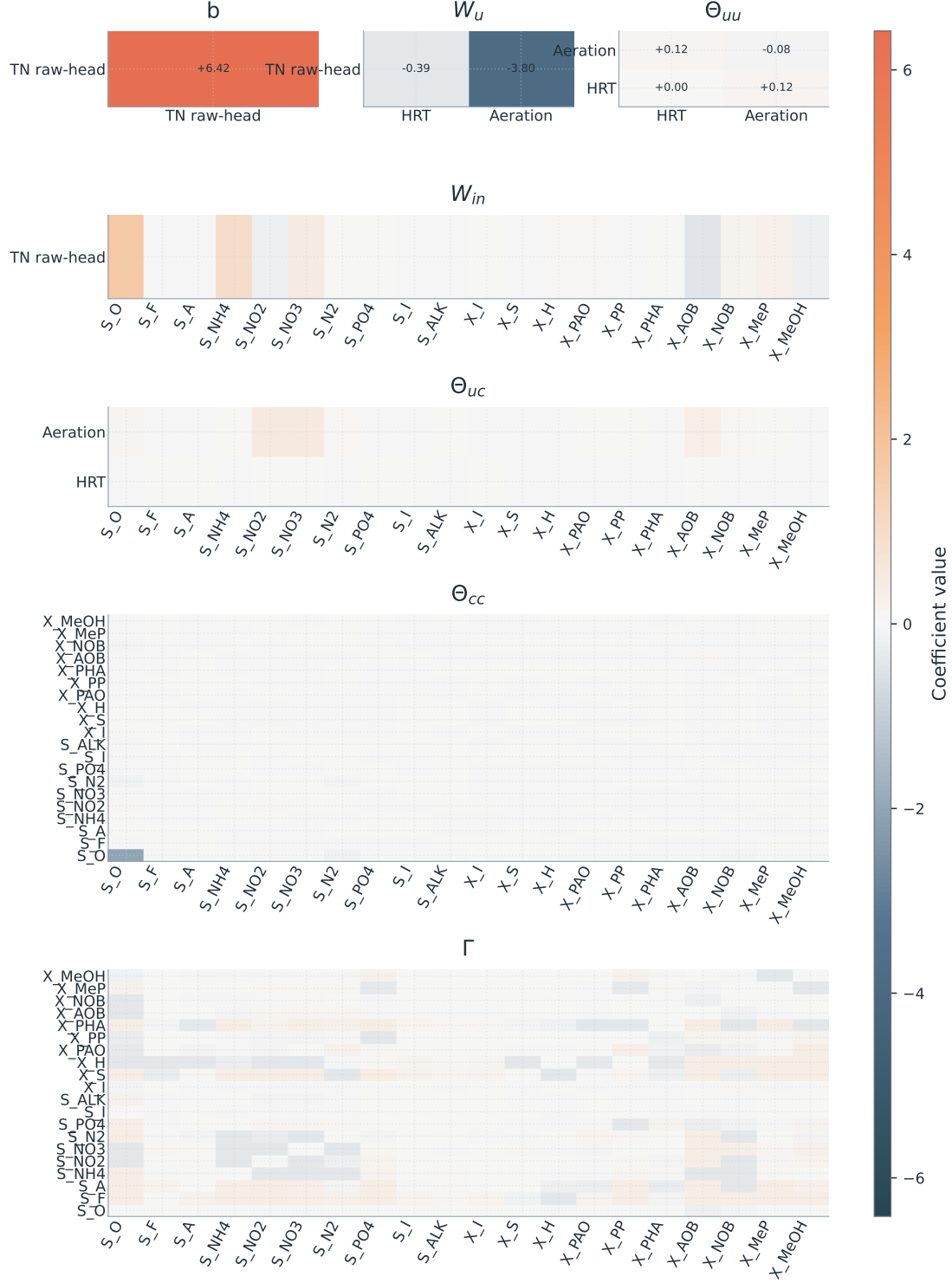


Figure S2: TN raw-head ICSOR coefficient atlas showing the reported-target blocks of $I_{comp}(I_F - \Gamma)^{-1}B$ together with the shared component-space coupling matrix Γ . The symmetric Θ_{uu} and Θ_{cc} panels display mirrored off-diagonal cells.

TP raw-head-specific ICSOR coefficient atlas

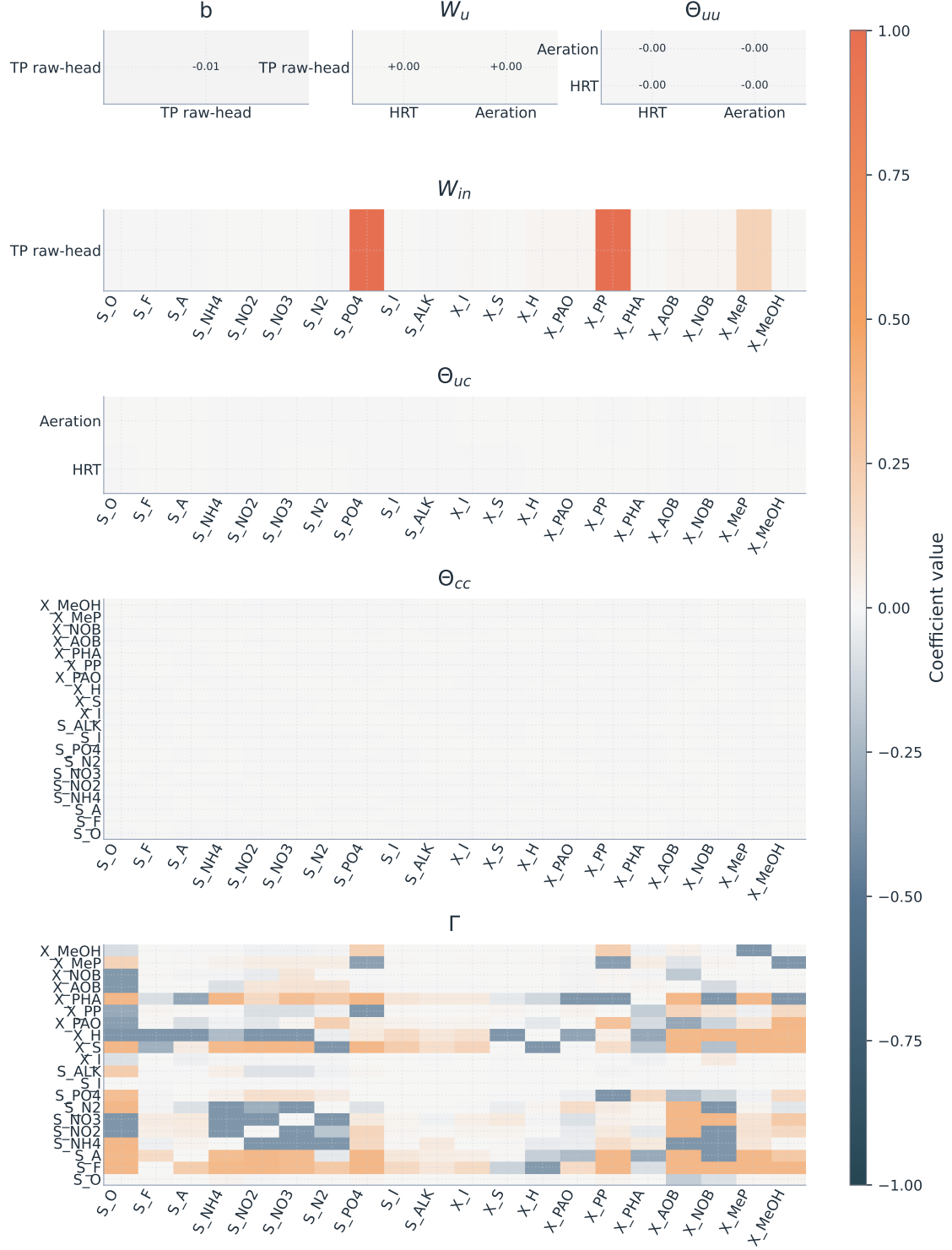


Figure S3: TP raw-head ICSOR coefficient atlas showing the reported-target blocks of $I_{comp}(I_F - \Gamma)^{-1}B$ together with the shared component-space coupling matrix Γ . The symmetric Θ_{uu} and Θ_{cc} panels display mirrored off-diagonal cells.

TSS raw-head-specific ICSOR coefficient atlas

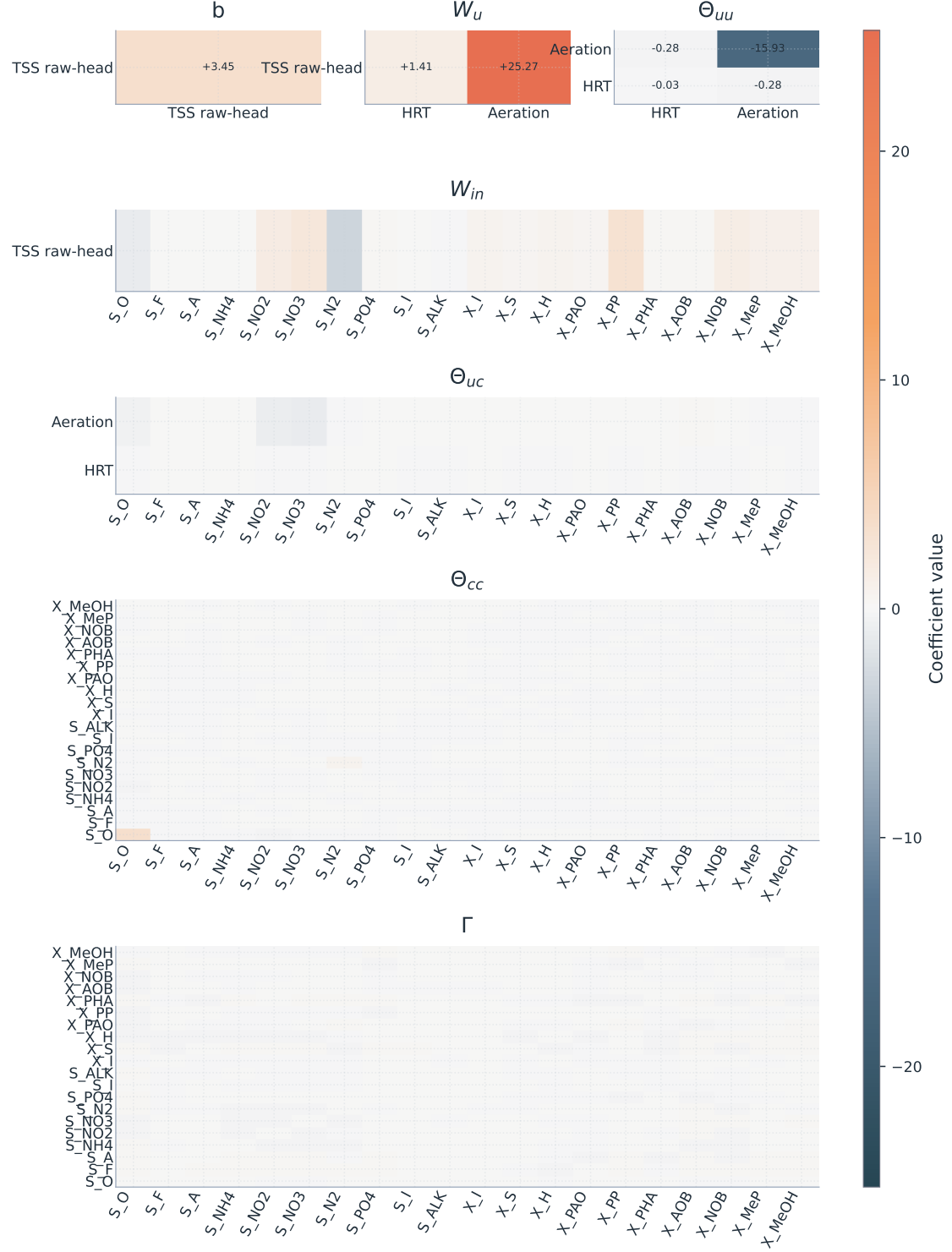


Figure S4: TSS raw-head ICSOR coefficient atlas showing the reported-target blocks of $I_{comp}(I_F - \Gamma)^{-1}B$ together with the shared component-space coupling matrix Γ . The symmetric Θ_{uu} and Θ_{cc} panels display mirrored off-diagonal cells.